



COMSATS University Islamabad

Department of Computer Science

Course Syllabus

Course Information

Course Code & Title: **CSC101 - Applications of Information and Communication Technologies**

Credit Hours: **3(2,1)**

Lecture Hours/Week: **2**

Lab Hours/Week: **3**

Pre-Requisites: **None**

Catalogue Description:

This course covers the basics of Information and Communications Technologies: Topics include: Overview of ICT; Computing Models; Computing Devices & their Types; Computer Systems & Components; Number Systems & Computer Codes; System & Application Software; Introduction to Databases & Information Systems; Computer Networks & Internet; E-Commerce; Security & Privacy; Intellectual Property Rights; ICT Applications: Health Industry, Banking, Expert Systems, and Digital Currencies; Emerging Trends in ICT: Cloud Computing, IoT, Virtual & Augmented Reality, Uses of Artificial Intelligence; and Problem Solving using Computer Programming.

Text and Reference Books

Textbooks:

1. Computer Science Illuminated, Nell Dale and John Lewis, Jones & Bartlett Learning, LLC, 2020
2. Understanding Computers: Today and Tomorrow, Comprehensive, Deborah Morley, Charles S. Parker, Cengage Learning, 2017.
3. Python Basics: A Practical Introduction to Python 3, David Amos, Dan Bader, Joanna Jablonski, and Fletcher Heisler, Real Python, 2021
4. Python Programming an Introduction to Computer Science, John M. Zelle, Franklin, Beedle & Associates Inc, 2016.

Reference Books:

5. Foundations of Computer Science, Forouzan, B., McGraw-Hill, 2017.
6. Problem Solving & Programming, Sprankle, M., Hubbard, J., Prentice Hall, 2012.

Week wise Plan:

Lecture #	CDF Unit #	Topics Covered	Reading Material
1.	1	Basic Definition of Information Technology (IT), Computer Technology & Communications Technology, Role of IT in Society, Computing Models: Turing Model, Von Neumann Model; Input Devices: Keyboard, Pointing & Touch Devices, Game Controllers, Optical Input Devices, Audio Visual Devices; Output Devices: Monitors, Audio Output, and Printers & Plotters.	Morley: Ch1, Ch4 Dale: Ch1
2.	1	How Computer Process Data; Central Processing Unit: Control Unit, Arithmetic & Logic Unit, System Clock & Machine Cycle; Memory: Volatile & Non-Volatile, Flash Memory, Registers, Cache Memory; Bus & Types, and Ports.	Morley: Ch2 Dale: Ch2, Ch3
3.	1	Storage Devices: Magnetic Storage Devices, Optical Storage Devices, Solid State Storage Devices, and Data Read & Write Process.	Morley: Ch3
4.	1	How Computer Represent Data: Number Systems, Bits & Bytes, and Text Codes.	Morley: Ch2
5.	1	Number Systems: Binary, Octal, Hexadecimal, and Conversions.	Morley: Ch2

6.	1	Number Systems Arithmetic: Addition & Subtraction of Decimal, Binary, Octal, and Hexadecimal Number System.	Dale: Ch2
7.	2	Computer Software: System Software, Application Software, Operating System (OS): Basic Purpose of OS, Types of OS, Operating System Functions: Providing User Interface, Running Programs, Managing Hardware, Utility Programs, PC Operating System, Network Operating System, and Embedded Operating System.	Morley: Ch5, Ch6 Dale: Ch10
8.	2	Database: What is a Database? Advantages & Disadvantages of the DBMS Approach, Data Concepts & Characteristics, Data Definition, Data Hierarchy Entities & Entity Relationships, Data Integrity, Security & Privacy, and Centralized vs. Distributed Database Systems.	Morley: Ch14
9.	2	Networking Concepts; Networking Applications: Telecom Industry, Videoconferencing; Network Characteristics: Types of networks, Network Topologies, Network Architectures, and Network Size & Coverage Area.	Morley: Ch7
10.	2	Data Transmission Characteristics: Bandwidth, Analog vs. Digital Signals, Transmission Type & Timing, Delivery Method; Networking Media: Wired Networking Media & Wireless Networking Media; Communications Protocols & Networking Standards; and Networking Hardware.	Morley: Ch7, Ch8
11.	2	Basic Security Concepts: Threats to User: Identity Theft, Loss of Privacy, Online Spying Tools, Online Theft, Online Fraud, & Other Dot Cons; Threats to Hardware: Power Related Threats, Hardware Loss, Hardware Damage, System Failure, Unauthorized Access & Unauthorized Use, Natural Disasters; Threats to Data: Malware, Virus & Malicious Program, Cybercrime, Cyber-Terrorism, and Counter Measures.	Morley: Ch9
12.	2	Information Systems & System Development: Need for System Development Enterprise Architecture, Business Intelligence (BI), Users of Information Systems; Types of Information Systems; Responsibility for System Development; System Development Life Cycle (SDLC); and Approaches to System Development.	Morley: Ch12 Dale: Ch12
13.	2	E-commerce: Overview, E-commerce Basics, Business Applications, Global Trends; Impacts: Impacts on Market & Retailers, Impact on Supply Chain Management, Impact on Employment, Impact on Customers, Impact on Environment, Impact on Traditional Retail; and Distribution Channels.	Morley: Ch11
14.	2	Computer System Security and Privacy	Morley: Ch15 Dale: Ch17
15.	2	Intellectual Property Rights, Ethics, Health Access, and the Environment	Morley: Ch16
16.	Mid Term Exam		
17.			
18.	3	ICT Applications: Health Industry, Banking, Expert Systems, and Digital Currencies	Dale: Ch14
19.	3	Cloud Computing: Overview, Advantages & Disadvantages, Characteristics; Service Models; Deployment Models; and Security & Privacy.	Dale: Ch14
20.	3	IoT: Applications; Trends, Characteristics & IoT Adoption Barrier	Dale: Ch14

21.	3	Augmented Reality (AR): Technology, Possible Applications, Remote Collaboration, and Danger of AR	Dale: Ch14
22.	3	Artificial Intelligence: Uses of AI, and Applications	Dale: Ch13
23.	4	General Problem Solving with Sequential Structure , Discussion & Practice using Pseudocodes, and Generating Flowcharts.	Dale: Ch6
24.	4	Problem Solving with Decision Structure , Discussion & Practice using Pseudocodes, and Generating Flowcharts.	Dale: Ch6
25.	4	Problem Solving with Repetition Structure , Discussion & Practice using Pseudocodes, and Generating Flowcharts.	Dale: Ch7
26.	4	Problem Solving with Procedure, Functions & Lists , Discussion & Practice using Pseudocodes, and Generating Flowcharts.	Dale: Ch7
27.	4	What is a Computer Program? Introduction to Python, Indentation, Control Structures, and First Python Program.	Amos: Ch3
28.	4	Variables: Integer Values, Variables & Assignment, Identifiers, Floating-Point Values, User Input, eval Function, Controlling print Function, and Difference between Expressions & Statements in Python.	Amos: Ch5
29.	4	Operators & Operands; Variable Comparison, Logical & Boolean Operations, Conditional Statements, and Order of Precedence. Nested Conditionals, Multi-way Decision Statements, Conditional Expressions, Errors in Conditional Statements, and else-if statement vs if-if Statement.	Amos: Ch8
30.	4	Strings, Control Codes within Strings, Slicing & Indexing Strings, Concatenation, and Arithmetic Operators with Strings.	Amos: Ch4
31.	4	While Loop, Conditionals within a Loop, Counting using while Loop, and break & continue Statements. Nesting while Loop, Nesting for Loop, Nesting while inside for Loop, and Nesting for Inside while Loop. For Loop, Conditionals within a Loop, Counting Using for Loop, and in Operator with for Loop.	Amos: Ch6, Ref. Material
32.	4	Functions: Predefined vs User Defined Function, Using Predefined Functions, Standard Mathematical Functions, and Random Numbers. User Defined Functions: Function Basics, Defining Functions, Calling Functions, Defining Main Function, and Parameter Passing. Void Function vs Value Returning Function,	Amos: Ch6, Ref. Material

Final Term Exam

Graduate Attributes (GA)

S.#	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify, formulate, research literature, and solve <i>complex</i> computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
5	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to <i>complex</i> computing activities, with an understanding of the limitations

Course Learning Outcomes (CLOs)

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA
CLO's for Theory				

CLO-1	1	Explain the basic computing models and related hardware.	<i>Understanding</i>	2
CLO-2	2	Work out with different number systems and codes.	<i>Applying</i>	2,3
CLO-3	3-4	Describe the fundamental concepts of ICT domains.	<i>Understanding</i>	2
CLO-4	5	Solve computing problems using problem-solving process.	<i>Applying</i>	2,3,5
CLO-5	6	Describe the concepts of variables, conditional, repetitive structures, and functions using a programming language.	<i>Understanding</i>	2,3
CLO's for Lab				
CLO-6	3-5	Demonstrate the usage of productivity software, problem solving tool, and web technology by performing appropriate tasks.	<i>Applying</i>	5
CLO-7	6	Write a program in a programming language to solve a simple problem.	<i>Applying</i>	2-5

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6	CLO-7
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-	-
Assignments	-	-	Assignment 1&2	Assignment 3	Assignment 4	Lab Assignment 1-3	Lab Assignment 4
Mid Term Exam	Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-	-	-
Final Term Exam	Final Term Exam					-	-
Project	-	-	-	-	-	-	Lab Project

Policy & Procedures

- **Attendance Policy:** Every student must attend 80% of the lectures as well as laboratory in this course. The students falling short of required percentage of attendance of lectures/laboratory work, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Total
Theory (T)	15	10	25	50	100
Lab (L)	-	25	25	50	100
Final Marks (T+L)	$(T/100) * 67 + (L/100) * 33$				

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Marks	>= 85	80 - 84	75 - 79	71 - 74	68 - 70	64 - 67	61 - 63	58 - 60	54 - 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.
- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.
- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.